

NEXUM SUUM INCORPORATED

24-Hour Boiler System SOP

These SOPs apply to continuous boiler operations for major manufacturers such as Aerco, Bryan, Cleaver-Brooks, Johnston, and Fulton. Tasks are divided by shift and align with safety and performance standards.

Afternoon Shift (2 PM - 10 PM)

- Confirm control panel readings match gauges - Clean air intake filters (Aerco, Cleaver-Brooks models) - Record flue gas analyzer readings if available - Manually test emergency shutdown - Inspect chemical injection flow and treatment pump - Check economizer or heat recovery temps (if applicable)

Night Shift (10 PM - 6 AM)

- *Cycle blowdown (if automatic) and check for flashback - Ensure feedwater temperature and deaerator levels are stable - Inspect combustion air damper operation - Review boiler runtime hours and backup boiler readiness - Ensure lag boilers are prepped and controls are functional - Perform flashlight check for carbon buildup or hot spots*

Checklist: Morning Shift (6 AM - 2 PM)

Log boiler pressure, steam output, and stack temp	Yes / No
Conduct visual inspection of burner flame via sight port	Yes / No
Check water level (sight glass or remote sensors)	Yes / No
Verify all safeties (LWCO, high-limit, flame scanner status)	Yes / No
Ensure blowdown is scheduled and noted	Yes / No
Check for leaks around feedwater piping, valves, and traps	Yes / No
Calibrate O ₂ sensor or note deviation	Yes / No